

Reticular Local Abstraction and Compact Reticular Computability

An Epistemic Position Paper on Multi-Level Models,
Epistemic Neural Architectures, and Topological Proto-Consciousness

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Abstract

Classical computability theory provides a complete and elegant account of what is computable at a single level of description, but it is largely silent about how computation proceeds in multi-level scientific systems characterized by coarse-graining, information collapse, and emergent macro-phenomena. Modern sciences—from systems biology to condensed matter physics and neural AI—routinely work with chains of models spanning distinct levels, where injective, quasi-injective, and non-injective relations coexist.

This article is an explicitly *epistemic position paper*. It introduces *Reticular Local Abstraction* (RLA) and *Compact Reticular Computability* (CRC) as a conceptual framework for thinking about multi-level *models* and epistemic architectures, not about the ultimate ontology of the world. Scientific domains, and the formalisms used to model them, are classified along a spectrum of Turing-likeness not as claims about reality itself, but as tools for discretising and organising human experience and measurement. RLA treats scientific levels as pairs of state spaces and rule sets within such modelling practices, linked by transmissions that may preserve distinctions or collapse them. CRC characterises when such a multi-level structure is (i) *internally ontologically autonomous* (in a model-internal sense), (ii) *epistemically closed*, and (iii) fully *Turing-computable*, all relative to a chosen epistemic horizon. Under these conditions, undecidability and emergence become structural properties of the *modelling reticulum*: limits can propagate along quasi-injective links, while non-injective collapses generate novel macro-properties that are nonetheless computable at their level.

A proof-of-concept case study on a generalist bryophyte shows how equations, parameters, and feedbacks from the relevant literature can be assembled into a compact reticulum within the bryophyte’s internal ontological horizon as reconstructed by plant physiology and ecology (Annex B). In such a setting, and provided the embedded science is sufficiently expressive and free of major error, one can reasonably expect synthetic observations to approximate epistemic equivalence to empirical ones in blind assessment by domain experts; at present this remains a design goal rather than a validated result. We then move from this natural CRC to engineered CRC architectures: convolutional neural networks (CNNs) are reinterpreted as RLA reticula, and *Epistemic CNNs* (ECNNs) extend them with an epistemic head capable of emitting unknown and contradiction under a falsifiable protocol.

To implement epistemic closure in practice, we propose *Epistemic Computational Units* (ECU), realised as large language models constrained by deterministic epistemic matrices, and *Epistemic Computational Entities* (UCE) arranged in functional rings. Because each ECU can, under matrix control, generate structured epistemic objects—such as coarse-grained models of prefrontal functions, abstract decision policies, or even Jungian-style archetypal patterns—in a way that is conditionally dependent on the outputs of other ECUs, the relevant scale for proto-conscious behaviour is not the raw number of low-level neurons or parameters. What matters is the topology and closure of the reticulum of such high-level epistemic objects, including cases where the objects model “alien” mechanisms outside familiar human phenomenology.

Finally, we outline—as a clearly marked speculative perspective—a topological conjecture of proto-consciousness in compact epistemic reticula: consciousness is treated not as a scalar function of size, but as a property of reticular configuration, closure, and computable self-limits within a modelling perspective. The bridge to the Principle of Computational Equivalence (PCE) and the conjectural aspects of proto-consciousness are confined to a short section in this paper and developed in detail only in Annex D. Formal definitions, proofs, and technical details for RLA, CRC, ECNN, ECU/UCE, and the PCE bridge are provided in the accompanying annexes A–E, to which this paper serves as an overarching conceptual manifesto.

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1 Introduction: Scope and Epistemic Stance

1.1 Classical computability and its domain of adequacy

Classical computability theory, from Turing’s analysis of effective procedures (Turing, 1936) and Church’s lambda calculus (Church, 1936) to Rice’s theorem on extensional properties of programs (Rice, 1953), provides a robust and widely accepted characterisation of which functions on discrete domains are computable. Within this framework, a system is typically captured by a single-level formalism (e.g., a Turing machine, a register machine, or a cellular automaton), and questions of decidability or undecidability are posed within that level.

This perspective is powerful and complete for its intended scope: reasoning about algorithms and decision problems defined on a fixed configuration space. However, it is largely indifferent to how scientists actually organise knowledge across *multiple* levels of description in empirical domains, and to the systematic use of coarse-graining, aggregation, and information loss in building tractable models.

1.2 The multi-level nature of scientific practice

In contemporary science, few domains are genuinely mono-level. Systems biology links molecular dynamics, cellular regulation, tissue physiology, and organismal phenotypes (Noble, 2012). Condensed matter physics moves between microstate ensembles and emergent phases of matter (Laughlin & Pines, 2000). Neuroscience and cognitive science work with neuronal circuits, functional networks, and behavioural-level descriptions (Bechtel & Abrahamsen, 2005). In each case, scientific practice does not simply stack models: it constructs chains of levels with specific relations among them.

These relations are not always information-preserving. Many modelling moves amount to deliberate *collapses*: coarse-graining over microstates, ignoring fine distinctions, or aggregating variables into composite indices. At the same time, theoretical computer science tells us that any Turing-like formalism capable of universal computation inevitably hosts undecidable properties (Chaitin, 2005). The interaction between these two facts—multi-level modelling via information collapse, and intrinsic limits of computation—is not explicitly addressed in classical computability.

1.3 Epistemic, not ontological, ambitions

The framework developed in this paper is explicitly *epistemic*. The levels, transmissions, and reticula we discuss are not presented as literal strata of an objective, mind-independent ontology. Rather, they are structures *within scientific modelling practice*: ways of discretising and organising experience, measurement, and theory.

When we classify a scientific domain as “Turing-like” or “non-Turing”, we do not attribute computational properties to reality itself. We classify the *formalisms* currently used to represent and predict phenomena in that domain along a spectrum of Turing-likeness. Similarly, when we speak of “ontological independence” or an “internal ontological horizon”, we mean the internal ontology induced by a modelling framework: the set of entities, laws, and relations that the model takes as its explanatory inventory. We remain neutral on the metaphysics of the world; our claims concern the structure and limits of multi-level models.

This stance is close to model-based and structural approaches in philosophy of science, where models are understood as representational artefacts rather than mirrors of nature

(e.g., Humphreys, 2016). RLA and CRC are proposed as tools to reason about such models and their interrelations, not as ontological doctrines.

1.4 Type and scope of the contribution

This article should therefore be read as a *position paper* or conceptual manifesto. Its contributions are:

1. A conceptual and partially formal framework (RLA/CRC) for reasoning about multi-level models and their computability.
2. A proof-of-concept natural case study (a bryophyte reticulum) illustrating how CRC can be instantiated in an empirical domain.
3. A proposal for engineered CRC architectures in neural AI (ECNNs and ECU/UCE rings), positioned with respect to existing work on uncertainty, abstention, and calibration.
4. A clearly demarcated speculative perspective on proto-consciousness as a topological property of compact epistemic reticula.

More concretely, the main body of the paper has two modest goals:

- to propose RLA/CRC as a multi-level grammar and to illustrate it via the bryophyte reticulum;
- to show ECNNs and ECU/UCE rings as examples of engineered CRC architectures.

The bridge to the Principle of Computational Equivalence and the topological conjecture of proto-consciousness are intentionally confined to a short, clearly marked section (Section 8) and to Annex D. They should be read as speculative extrapolations rather than as core technical claims.

Detailed technical developments and proofs are contained in separate documents, referred to here as: Annex A (*Foundations of Reticular Local Abstraction and Compact Reticular Computability*), Annex B (*Compact RLA Topology for a Generalist Bryophyte*), Annex C (*Epistemic CNNs: Formalisation, Training, and Metrics*), Annex D (*Reticular Local Abstraction, Epistemic CNNs, and the Bridge to the Principle of Computational Equivalence*), and Annex E (*Epistemic Computational Units and Entities: ECU/UCE Design and Protocols*).

How to read this package

Different readers may reasonably follow different paths through the material. A minimal “reading strategy” is:

- **Philosophy / theory of science:** main paper + Annex A + Annex D.
- **Machine learning / AI:** main paper + Annex C + Annex E.
- **Ecology / biology:** main paper + Annex B.

Readers interested only in the speculative bridge to proto-consciousness can focus on Section 8 and the corresponding parts of Annex D.

Methodological note on AI-assisted drafting

The conceptual development, formalisation and literature analysis reported in this work were conducted by the human author, who retains full scientific responsibility for all claims, interpretations and conjectures. During the drafting process, the author employed several AI-based assistants (large language model systems configured with different roles and focuses) as methodological tools for:

- supporting the identification and preliminary organisation of relevant literature, on the basis of queries and selection criteria defined by the author;
- assisting in the production, restructuring and linguistic refinement of draft texts, always under the direct supervision and editorial control of the author;
- exploring alternative formulations, notational variants and local consistency checks across the RLA/CRC, ECNN, ECU/UCE and PCE frameworks.

Given the overall breadth and multidisciplinary nature of the project, the resulting bibliography is necessarily extensive and spans heterogeneous domains (computability theory, philosophy of science, complex systems, ecology, neuroscience, machine learning, and AI governance). All cited references have been manually checked by the author for existence, authorship, basic content and thematic relevance: no intentionally fictitious work is referenced. However, in view of the scale and heterogeneity of the material, it cannot be guaranteed that every micro-level reconstruction or bibliographic linkage is free from error or artefacts typical of AI-based drafting (“hallucination” in the sense of overconfident but inaccurate detail).

This caveat is particularly relevant for the scientific supply chain underlying the bryophyte case study and its compact reticulum (Annex B). The domain literature on moss physiology and ecology is real and has been selected and cross-checked by the author, but some specific technical details—for example, precise functional forms or parameterisations of individual equations when only secondary descriptions or paywalled sources were accessible—may incorporate approximations or inferred structures that the author is not in a position to verify line by line against the original publications. Where such discrepancies exist, they should be interpreted as local modelling approximations rather than as deliberate misrepresentation of the underlying science.

In the author’s assessment, these possible fine-grained inaccuracies do not materially affect the intrinsic value of the work: neither the definition of the RLA/CRC framework nor the construction of the compact bryophyte reticulum, whose role in the paper is that of a proof-of-concept epistemic object. The conceptual claims, structural results and conjectures advanced here remain independent of any single equation or parameter choice in the case study. The use of AI tools is therefore to be understood strictly as a form of assisted drafting and semantic support, and not as a delegation of scientific authorship, judgement or responsibility.

2 Reticular Local Abstraction: A Multi-Level Grammar

2.1 Levels as epistemic slices of practice

Reticular Local Abstraction (RLA) starts from the observation that scientific disciplines carve reality into *levels* of description, each with its own observables and operative rules. In our framework, these levels are taken to be *epistemic slices*: stable patterns in how communities measure, represent, and reason about phenomena, not ontological layers of being.

Formally, a level is represented as:

$$L = \langle D(L), \Sigma(L) \rangle,$$

where $D(L)$ is a space of observable or representable states (e.g., concentration vectors, population densities, firing rates) and $\Sigma(L)$ is a set of rules or operations that generate new observations or govern dynamics within that domain.

RLA does not prescribe what the “right” levels of nature are; it provides a language to talk about how scientific practice organises levels and relates them.

2.2 Transmissions: injective, quasi-injective, and non-injective

Between two adjacent levels L_i and L_{i+1} , RLA posits a *transmission*:

$$\tau_{i \rightarrow i+1} : D(L_i) \rightarrow \mathcal{P}(D(L_{i+1})),$$

which maps states at level L_i to sets of states at level L_{i+1} within the modelling context. Transmissions can be categorised as:

- **Injective**: preserve all distinctions within the relevant epistemic domain; if $x_1 \neq x_2$ in $D(L_i)$, their images remain disjoint at the level of representation.
- **Quasi-injective**: preserve distinctions on a subset $C_i \subseteq D(L_i)$ of *critical states*, e.g., those that carry Turing-level computational structure or are deemed relevant by the domain.
- **Non-injective**: implement a *collapse* C , mapping many distinct states to fewer equivalence classes (coarse-graining, saturations, thresholds), as defined by modelling choices.

These categories capture the basic ways information can travel across levels in scientific representations: preserved, partially preserved, or collapsed.

2.3 Turing-like and non-Turing regimes as properties of formalisms

RLA also distinguishes between levels according to the computational regime of the formalisms used to model them:

- **Turing-like**: levels represented by formalisms that can, in principle, simulate universal computation (e.g., certain dynamical systems, sufficiently expressive programming languages, or neural architectures).
- **Turing-in-potential**: levels represented by formalisms that approximate universality but are constrained by precision, memory, or energy.

- **Non-Turing:** levels represented by formalisms constrained to weaker forms of computation (e.g., linear PDEs solved with fixed schemes).

These labels attach to the *representational tools* rather than to nature itself. Undecidability theorems apply to Turing-like formalisms; RLA tracks how the corresponding limits interact with transmissions among levels in a multi-model chain.

2.4 Sample formalisation

For clarity, we extract one of the central ideas into a formal statement. Full development is provided in Annex A.

Epistemic reticular system. An *epistemic reticular system* is a finite sequence $R = (L_1, \dots, L_n)$ of levels $L_i = \langle D(L_i), \Sigma(L_i) \rangle$ together with transmissions $\tau_{i \rightarrow i+1} : D(L_i) \rightarrow D(L_{i+1})$ (we restrict to single-valued mappings for simplicity) such that each $\Sigma(L_i)$ and $\tau_{i \rightarrow i+1}$ is effectively computable by some Turing machine.

Critical set. A subset $C_i \subseteq D(L_i)$ is *critical* if the restriction of $\Sigma(L_i)$ to C_i is Turing-complete in the sense that it can simulate any Turing machine up to a computable encoding.

Theorem (Propagation of undecidability along quasi-injective transmissions). Let R be an epistemic reticular system, and let $C_i \subseteq D(L_i)$ be a critical set. Suppose $\tau_{i \rightarrow i+1}$ is injective on C_i and effectively computable. Then for any non-trivial extensional property $U \subseteq C_i$ that is undecidable with respect to $\Sigma(L_i)$, the induced property

$$U' := \{y \in \tau_{i \rightarrow i+1}(C_i) \mid y = \tau_{i \rightarrow i+1}(x) \text{ for some } x \in U\}$$

is undecidable with respect to the induced dynamics on $\tau_{i \rightarrow i+1}(C_i)$.

The intuition is straightforward: if τ is injective on C_i , then deciding U' at level L_{i+1} would amount to deciding U at level L_i , contradicting undecidability. The formal proof reduces to Rice-style arguments and is given in Annex A. Analogous results hold for quasi-injective transmissions on restricted domains.

As an informal corollary, in any CRC reticulum where a Turing-complete micro-level is connected to a macro-level via a quasi-injective transmission on the critical subset, no macro-level compression can decide non-trivial extensional properties of the underlying micro-dynamics without violating computability theory. In the bryophyte reticulum, this means that undecidable properties of a sufficiently expressive gene-regulatory model cannot be made decidable simply by passing to morphological or ecological aggregates; in ECNNs, undecidable properties of internal feature dynamics cannot be eliminated by adding further deterministic heads.

Similarly, one can prove that non-injective transmissions generically support emergent properties not definable at lower levels (Proposition 2 in Annex A), grounding a corresponding emergence axiom in a more classical representational setting.

2.5 Operational metrics

To move beyond qualitative claims, RLA introduces metrics such as a collapse coefficient $CC(\tau; S)$ for a finite set $S \subseteq D(L_i)$, an index of emergence (IE) counting non-definable macro-properties, and an irreducibility index (RIR) inspired by computational mechanics. These quantify how strongly a transmission collapses distinctions and how much macro-level organisation resists compression in the modelling chain. Details and examples are provided in Annex A.

3 Compact Reticular Computability

3.1 Compact reticula: internal ontology, closure, and computability

Given a chain of levels and transmissions, RLA alone does not yet tell us when the entire structure of models can be treated as a well-posed computational object. For this we introduce the notion of a *compact reticulum*, defined by three conditions:

- **Internal Ontological Autonomy (IOA).** Explanatory entities and laws are internal to the reticulum *relative to a chosen domain of discourse*; external inputs are filtered through non-injective boundary mappings. There is no appeal to ad hoc external entities that function as oracles within the model. This is a condition on the *self-sufficiency* of the modelling ontology, not on the metaphysics of the world.
- **Epistemic Closure (EC).** All phenomena of interest, as defined by the modelling agenda, are explainable by internal levels and transmissions. No appeal to meta-levels outside the reticulum is required to account for these phenomena.
- **Turing-Computability (TC).** All rules in $\Sigma(L_i)$ and transmissions $\tau_{i \rightarrow j}$ are algorithmically implementable and simulable by a Turing machine (subject to usual resource constraints).

When IOA, EC, and TC hold jointly, we say the reticulum is *compact* in the epistemic sense: it forms a self-contained computational representation of a domain.

3.2 Definition: Compact Reticular Computability (CRC)

We can now extend the notion of computability from single-level systems to multi-level reticula of models:

Definition (Compact Reticular Computability). A multi-level system $R = \{L_1, \dots, L_n\}$ with transmissions $\{\tau_{i \rightarrow i+1}\}$ is *compactly reticularly computable* if:

1. R forms a compact reticulum (satisfying IOA, EC, TC) relative to a specified epistemic horizon.
2. There exists a Turing machine T that simulates every $\Sigma(L_i)$ and $\tau_{i \rightarrow i+1}$ to arbitrary precision.
3. There exist macro-properties at higher levels L_j that are not definable as predicates on $D(L_i)$ for $i < j$, yet are computable *at their level* under $\Sigma(L_j)$.
4. Undecidability constraints originating in Turing-like levels can either propagate along quasi-injective chains (Theorem above) or be blocked by non-injective collapses (Proposition 2 in Annex A), without invoking non-computable mechanisms.

CRC thus characterises when a chain of models can be treated as a unified computational object, with internal limits and emergent properties that are themselves features of the modelling structure.

3.3 Consequences: co-existence of limits and emergence

Within CRC, undecidability and emergence are no longer in tension:

- Along chains of quasi-injective transmissions, undecidable properties propagate, in the sense that no higher-level compression in the modelling chain can decide them without contradicting computability theory.
- Where non-injective collapses act on relevant subsets, undecidability can be effectively “blocked” for certain questions (they become ill-posed or undefined at higher levels), but new macro-properties arise that are not reducible to micro-level predicates within the model.

This provides a unified lens on classical debates about reductionism and emergence in complex systems (Anderson, 1972; Humphreys, 2016; Kauffman, 2000), understood as debates about the structure and limits of multi-model representations rather than about the world’s ontology.

3.4 Relation to classical computability

CRC does not modify the classical definition of computability for single functions or machines. Rather:

- Every Turing machine can be trivially seen as a degenerate one-level reticulum.
- CRC adds structure on top of standard machines by describing how multiple computable levels, connected by injective/quasi-injective/non-injective transmissions, behave as a whole as a representation.

Classical computability thus appears as the special case of CRC where there is only one level and no information collapse in the modelling chain. Formal discussion and further theorems are deferred to Annex A.

4 Case Study: A Compact Reticular System for a Generalist Bryophyte

4.1 Motivation and scope

To show that CRC is not a purely abstract notion, we consider a case study: a compact reticular system modelling a generalist bryophyte (moss). The aim is to assemble existing equations, parameters, and feedbacks from the literature into a reticulum that satisfies IOA, EC, and TC *within the internal ontological horizon of the bryophyte as reconstructed by domain science*. The resulting reticulum is not claimed to mirror the organism’s true ontology, but to serve as an epistemic object that supports explanation and prediction.

The complete reticulum, with all parameters, equations, and validation procedures, is documented in Annex B.

4.2 Construction pipeline

The construction proceeds in five main steps:

1. **Collection of domain knowledge:** equations and parameters describing photochemistry (e.g., Michaelis–Menten kinetics), tissue hydraulics, gas exchange, energy balances, and population dynamics are drawn from established models in plant physiology and ecology.
2. **Stratification into levels:** variables are grouped into levels such as:
 - L_1 : genomic–epigenetic core,
 - L_2 : transcription/translation,
 - L_3 : metabolic regulation,
 - L_4 : cell physiology,
 - ...
 - L_{10} : macroclimate context,
 - L_{20} : meta-systemic health/resilience.
3. **Definition of transmissions:** saturations, thresholds, and operating windows define non-injective collapses; regions where distinctions matter define quasi-injective behaviour; some quantities are passed injectively.
4. **Parameter calibration:** local fits to experimental data (e.g., photosynthesis curves) and global fits to growth and stress-response time series ensure empirical adequacy at each level, as far as available data allow.
5. **Verification of compactness:** checks are made that all explanatory entities are internal (IOA), that observed variables are covered by internal levels (EC), and that all dynamics are algorithmically simulable (TC).

A schematic grouping of abstraction levels is summarised in Table 1; full details are given in Annex B.

Quantitatively, the reticulum includes on the order of 42 biophysical parameters, 38 dynamic equations, around 30 conditional actions, approximately 22 collapse–emergence metrics, and 2 composite indices, as detailed in Annex B.

4.3 Compactness in practice

Internal Ontological Autonomy. The bryophyte reticulum does not rely on external symbolic oracles; environmental drivers (e.g., temperature, light, rainfall) are treated as exogenous inputs that are collapsed into boundary conditions via non-injective mappings. All internal behaviour is generated by levels and transmissions that remain inside the defined epistemic horizon of the organism and its immediate environment.

Epistemic Closure. For the set of observables considered (growth curves, photosynthetic rates, resilience indices), explanations are provided by internal levels and their interactions. When the underlying domain science is sufficiently expressive and the modelling choices are adequate, the reticulum is intended to capture all relevant explanations for these observables without recourse to higher meta-levels.

Table 1: Schematic grouping of abstraction levels in the bryophyte reticulum (full specification in Annex B).

Group	Example levels	Description
Micro	L01–L04	Genomic and epigenetic states, transcription and translation, metabolic and hormonal regulation, cell physiology (ROS, photosynthesis, turgor).
Meso	L05–L10	Tissues and microstructures, morphogenesis and architecture, reproductive cycle, biotic interactions, immediate microenvironment, macroclimatic forcing.
Memory / homeostasis	L11–L16	Stress memory, systemic control and homeostasis, soil and substrate, endogenous rhythms, senescence and programmed cell death, intercellular signalling.
Macro	L17–L20	Geometric patterning, tissue mechanics, evolution and plasticity, systemic state and meta-observables (vitality, desiccation tolerance, composite health metrics).

Turing-Computability. All equations and update rules can be discretized and iterated algorithmically; the entire reticulum can be simulated by conventional numerical methods on digital hardware.

4.4 Synthetic observations and epistemic equivalence

When simulated under varying climatic drivers, the reticulum produces time series of growth, stress responses, and resilience indices. These *synthetic observations* live in the same space of variables as empirical observations and obey the same kinds of functional relationships, at least to the extent that the encoded science is correct.

Epistemic indistinguishability. Fix a set S of observables (e.g., subsets of humidity, vitality, ROS indicators, and composite health metrics) and an observer O (e.g., a plant ecophysicologist). A real trajectory $\mathbf{x}^{\text{real}}$ and a simulated trajectory $\mathbf{x}^{\text{sim}}$ over S are *epistemically indistinguishable* for O at threshold $\theta \in (0.5, 1)$ if, given only the outputs in S , O cannot assign with reliability $> \theta$ which trajectory is real and which is synthetic.

In this context, a natural working hypothesis is that, in a suitably designed blind test, a domain expert restricted to the conventional observational variables could find synthetic traces epistemically indistinguishable from real ones: the synthetic data would support the same explanations and decisions within the ontological horizon encoded in the reticulum. Whether this strong form of equivalence holds is ultimately an empirical question, dependent on the expressive adequacy of the science embedded in the model and on the absence of systematic errors. CRC provides the conceptual frame: if the model is compact and empirically adequate, then such epistemic equivalence is a principled design goal and evaluation criterion, not a metaphysical claim. At present, the bryophyte reticulum should be regarded as a *worked example* of CRC design rather than as a fully validated ecological model.

Conceptual blind test protocol. A minimal experimental design for testing epistemic indistinguishability might involve:

- *Displayed variables:* a fixed set S of observables, such as trajectories of water content,

vitality indices, ROS-related proxies, and composite resilience metrics, in standard graphical formats (time series, phase plots).

- *Experts*: a small panel of domain experts (e.g., 5–10 plant ecophysiologicalists) with experience in interpreting such data, but without access to the model internals.
- *Material*: a balanced set of real and synthetic trajectories over comparable time windows and environmental regimes, anonymised and randomly permuted.
- *Metric*: the fraction of cases in which experts can correctly label trajectories as “real” vs “synthetic” with reliability above θ ; epistemic indistinguishability corresponds to performance near chance, adjusted for any systematic biases.

This establishes a concrete example of CRC in a natural domain, which we now contrast with engineered CRC architectures.

5 From Natural CRC to Engineered CRC Architectures

5.1 Structural, not biological, generality

The bryophyte case illustrates that CRC can be realised in a biological context. However, nothing in the definition of CRC requires levels to be biological, chemical, or physical. The core ingredients are:

- discrete or discretizable state spaces $D(L_i)$,
- computable rule sets $\Sigma(L_i)$,
- transmissions with controlled injective/quasi-injective/non-injective behaviour,
- and satisfaction of IOA, EC, and TC relative to a chosen epistemic horizon.

CRC is thus *structural and epistemic*: it applies to any multi-level modelling architecture—natural or artificial—that can be organised into such a reticulum.

Crucially, the levels themselves can be chosen to represent high-level scientific or cognitive objects, provided they admit a computable representation. For example, one could define levels whose state spaces encode information typically associated with prefrontal cortical processing (e.g., working memory states, task sets), abstract psychological constructs such as Jungian archetypes, or even hypothetical “alien” mechanisms outside current human phenomenology, as long as they can be formalised as computational objects within an RLA reticulum. In this sense, CRC does not presuppose a low-level neuronal substrate; it concerns the organisation of epistemic objects, whatever their semantic origin.

5.2 Designing CRC in computational systems

This opens the possibility of *engineered CRC architectures*: computational systems deliberately designed to:

1. implement multiple levels of representation,
2. control how information is preserved or collapsed across levels,
3. make undecidability and uncertainty explicit where relevant,

4. and maintain epistemic closure within a definable operational horizon.

We now show that contemporary neural architectures, in particular convolutional neural networks (CNNs), already approximate this structure, and that with suitable extensions they can be turned into explicit CRC instances.

6 CNNs as RLA Reticula and ECNNs as Epistemic CRC Instances

6.1 CNNs through the lens of RLA

Convolutional neural networks, originally introduced for image and signal processing (LeCun et al., 2015), exhibit a multi-level architecture that maps naturally onto RLA notions:

- **Convolutional layers** perform local, shift-invariant feature extraction. For appropriate choices of kernels and non-linearities, these can be seen as *quasi-injective* on discriminative patterns: they preserve distinctions that matter for classification while discarding irrelevant noise.
- **Pooling layers** (max or average pooling) implement explicit *collapses*: multiple activations are aggregated into a single value, producing invariances (e.g., to small translations) at the cost of information loss.
- **Dense layers** at higher depth perform many-to-one mappings from rich feature sets to compact representations or logits, acting as semantic compaction layers.

Under this reading, a standard CNN is an RLA reticulum where $\tau_{i \rightarrow i+1}$ alternates between quasi-injective and non-injective behaviours, and where levels correspond to feature representations of increasing abstraction within the model.

6.2 Epistemic limitations of classical CNNs

Despite their success, classical CNNs treat uncertainty and epistemic limits implicitly:

- Outputs are often interpreted as calibrated probabilities, yet overconfidence and miscalibration are widely documented (Guo et al., 2017).
- There is no explicit mechanism to output “I do not know” or “this input contradicts my training assumptions”.
- Out-of-distribution (OOD) inputs can elicit confident but wrong predictions (Hendrycks & Gimpel, 2016).

In the RLA/CRC perspective, a CNN is a partially specified reticulum: it implements multi-level processing and collapses, but lacks an explicit *epistemic head* that would make ignorance, contradiction, and model limits *computable observables* of the architecture.

6.3 Related work on uncertainty, abstention, and calibration

There is a substantial literature on epistemic uncertainty in deep learning. Bayesian approaches approximate posterior uncertainty via variational methods or ensembles (Gal & Ghahramani, 2016; Lakshminarayanan et al., 2017). Selective classification and prediction with a reject option provide formal frameworks for abstaining on uncertain inputs (Chow, 1970; Geifman & El-Yaniv, 2017). Conformal prediction offers distribution-free methods for calibrated sets of predictions (Vovk et al., 2005). Calibration techniques aim to align predictive probabilities with empirical frequencies (Guo et al., 2017).

The ECNN proposal is not a replacement for these developments, but an attempt to integrate their core insights into a structural, reticular view. Rather than treating uncertainty, abstention, and calibration as add-on components, ECNNs embed them into an epistemic channel that is explicitly modelled as part of a multi-level reticulum.

6.4 Epistemic CNN (ECNN): adding an epistemic channel

An *Epistemic CNN* (ECNN) extends a CNN with an additional head L_{ep} that produces epistemic outputs, such as:

- a standard label prediction (or distribution over labels),
- an “unknown” state indicating rational rejection,
- a “contradiction” state indicating internal inconsistency.

Training is governed by an epistemic loss:

$$L = \alpha L_{\text{pred}} + \beta L_{\text{expl}} + \gamma L_{\text{meta}},$$

where:

- L_{pred} captures conventional predictive performance,
- L_{expl} penalises unstable explanations and miscalibration (e.g., via expected calibration error, Brier score),
- L_{meta} penalises confident errors and rewards correct unknown/contradiction decisions, especially on OOD or adversarial inputs.

This aligns with optimal reject rules (Chow, 1970), Bayesian approximations (Gal & Ghahramani, 2016; Lakshminarayanan et al., 2017), and selective classification frameworks (Geifman & El-Yaniv, 2017), but reinterprets them within a reticular, multi-level grammar.

6.5 ECNN as an engineered CRC instance

Viewed as a reticulum of representations, an ECNN satisfies CRC conditions:

- **IOA**: once trained, the network operates without external oracles; new inputs are processed through internal representations and the epistemic head.
- **EC**: explanations and limits are internalised in the feature hierarchy and the epistemic channel; the system can express when it lacks sufficient information.
- **TC**: the entire architecture is simulable by a Turing machine.

Moreover, non-injective pooling layers and dense compactions generate emergent invariants in higher-level features, while undecidability constraints (e.g., on properties of internal representations) either propagate or are blocked according to the structure of τ . Full technical specifications of ECNNs, loss components, and evaluation protocols (including Popper- χ challenge sets) are detailed in Annex C.

7 Epistemic Computational Units and Entities: LLM-Based Implementations

7.1 Why basic neurons are insufficient for epistemic closure

The epistemic head of an ECNN, as introduced above, is defined at the architectural level: it must evaluate coherence, detect conflict, and decide on unknown or contradiction. Implementing this head with a small feed-forward network or a single linear layer is generally insufficient to:

- interpret high-level features as structured propositions or evidential claims,
- cross-check them against rich domain constraints,
- generate explanations or rationales,
- handle complex OOD contexts in a principled way.

In CRC terms, a trivial head risks reducing the epistemic channel to a mere thresholding mechanism, undermining the goal of explicit, computable epistemic assessment within the model.

7.2 Epistemic matrices as constraints on LLMs

To address this, we introduce the notion of a *deterministic epistemic matrix* M . Operationally, M is a structured specification that includes:

- a set of admissible tasks (e.g., classification, consistency checking, evidence aggregation),
- an explicit list of allowed information sources and reference corpora,
- a grammar of permissible inference patterns and fallbacks,
- rejection and escalation policies (when to output unknown, when to flag contradiction),
- logging and audit requirements (trace formats, explanations, confidence tags).

M is not a matrix in the numerical sense but a rule- and schema-based object that can be implemented as a finite set of templates, validators, and control programs. It shapes how an LLM instance processes inputs and produces outputs, turning a general-purpose model into a bounded epistemic component.

7.3 Epistemic Computational Units (ECU)

Given an epistemic matrix M , we define:

Epistemic Computational Unit (ECU). An ECU is a session of a large language model L that operates under the control of an epistemic matrix M , such that:

- The input interface of the ECU is restricted to a fixed schema (e.g., tensors projected to symbolic descriptors, plus contextual metadata and epistemic tags).
- Internal prompts, tool calls, and decomposition strategies of L are constrained by M (allowed operations, allowed sources, required checks).
- The output interface is restricted to a finite ontology of epistemic outcomes: labels, unknown, contradiction, structured rationales, and suggested actions.
- All ECU behaviour is reproducible from M and the input, modulo the stochasticity explicitly allowed by M .

Conceptually, an ECU behaves as a “neuron” at the epistemic level: a unit of structured inference and self-limitation in the representational space. Unlike a scalar neuron, however, an ECU can generate complex, persistent epistemic objects, such as:

- structured summaries of information flow typically associated with prefrontal cortical regions (e.g., task sets, decision policies, conflict signals), when the underlying neuroscience is sufficiently formalised in an RLA reticulum;
- abstract psychological constructs (e.g., Jungian archetypes) represented as computable patterns of narratives, symbols, and behavioural expectations within a cultural state space;
- hypothetical or “alien” mechanisms not currently instantiated in human cognition, but defined by a formal theory that specifies their states, transitions, and observables.

Because ECUs can be conditioned on the outputs of other ECUs—via M imposing explicit dependencies and compositional prompts—they can build multi-step derivations in which high-level epistemic objects are refined, combined, or contradicted. In a reticular perspective, this conditional generativity is more relevant for proto-consciousness than the sheer count of low-level artificial neurons.

7.4 Epistemic Computational Entities (UCE) and functional rings

ECUs can be composed into *Epistemic Computational Entities* (UCE), arranged in functional rings inspired by cognitive architectures but defined purely in terms of information flow and evaluation:

- **Ring C0 (Executive/Planning):** normalises inputs from ECNNs into decisions or candidate plans, often by generating high-level policy objects.
- **Ring C1 (Valuation/Conflict):** evaluates costs, benefits, and conflicts, modulating C0 outputs and encoding them into decision-theoretic objects.
- **Ring C2 (Symbolic/Linguistic Memory):** retrieves and composes episodic and semantic context, including stable archetypal or conceptual structures.

- **Ring C3 (Metacognition/Justification):** checks global coherence, detects contradictions, and proposes unknown when warranted, often by constructing meta-representations of the system’s own reasoning trajectory.
- **Central Oracle:** integrates signals from rings, issues final outputs with epistemic status, and maintains a persistent internal narrative of decisions and limits.

All rings are composed of ECUs with their own matrices M , and the transmissions among rings form a compact reticulum (details in Annex E). The central oracle is *not* an oracle in the logical sense; it is simply the innermost UCE tasked with synthesis and self-limitation within the modelled decision process.

At this level, the basic “neurons” of the system are not activation scalars, but generative epistemic units producing and transforming rich objects.

7.5 Topological rather than scalar scaling

In many discussions of neural architectures and consciousness, “scale” is equated with the number of neurons or parameters: larger networks are assumed to be more powerful in a quasi-linear sense. The ECU/UCE construction shifts attention away from raw counts and towards topology:

- Each ECU is already a high-capacity epistemic unit that can generate structured objects (policies, internal narratives, archetypal patterns, scientific hypotheses) under deterministic constraints from M and conditioning on other ECUs.
- Increasing the number of ECUs beyond a certain point does not linearly increase functional richness; what matters is how they are *connected*: which transmissions are injective, which are collapsing, where feedback cycles and meta-level evaluations occur, and how complex epistemic objects are allowed to persist and interact.
- Two systems with similar parameter counts can differ radically in their epistemic properties, depending on whether they form a compact reticulum with computable self-limits and coherent high-level objects or a loosely coupled aggregate of uninterpreted activations.

In this sense, LLM-based ECUs serve to clarify that the relevant “neurons” for proto-consciousness are epistemic units in a reticular topology, not individual scalar units. Once the underlying science is expressive enough to define computable models of objects such as prefrontal information states, archetypal structures, or even “alien” mechanisms, the question of neuron count becomes largely orthogonal: the crucial feature is the organisation and closure of the network of such objects.

7.6 LLM as one realisation of ECU/UCE, not a conceptual prerequisite

ECNNs and the epistemic channel are defined at the architectural level; they do not logically require LLMs. In principle, any computable mechanism capable of rich inference and explanation under matrix M could serve as an ECU.

Large language models, however, provide a practically available implementation, offering:

- expressive power over natural and formal languages,

- the ability to integrate heterogeneous evidence,
- flexible explanation and meta-reasoning capabilities,
- and, crucially in this context, the ability to generate complex epistemic objects (narratives, internal policies, abstract constructs) in a way that can be strictly conditioned on other generated objects and controlled by M .

In this view, LLM-based ECUs/UCes are proposed as a *realisation* of engineered CRC architectures, and as a concrete illustration of how a topology of epistemic units operating over high-level representational objects can decouple proto-conscious properties from naive neuron-based scaling or strict biological parallelism. The same reticular principles would apply if, in the future, different computational substrates were able to instantiate equivalent epistemic matrices and object-generating capabilities.

8 A Topological Conjecture of Proto-Consciousness

8.1 Status of this section: speculative perspective

This section is intentionally speculative and may be read independently of the more technical proposals on RLA, CRC, ECNN, and UCE. The conjecture advanced here is not necessary for the rest of the framework; it is offered as a possible bridge between epistemic architectures and functional theories of minimal consciousness. The detailed bridge from CRC reticula to cellular automata and PCE is given only in Annex D.

8.2 Functional conditions for proto-consciousness

Building on CRC and UCE rings, we can pose, cautiously, a question about *functional proto-consciousness*: under what structural conditions might a computational system, considered as a modelling artefact, exhibit behaviours that are functionally analogous to minimal self-awareness, without assuming any phenomenological qualia?

We propose six minimal conditions:

1. **Computable self-limits**: the system can explicitly represent and emit unknown and contradiction when its internal evidence or coherence fails, under its epistemic matrix M .
2. **Internal telemetry**: it maintains a model of its own states and recent transitions, available for meta-evaluation.
3. **Controlled self-modification**: it can update internal parameters and rules within constraints imposed by M .
4. **Autonomous learning**: it can adjust its behaviour based on experience, within its epistemic horizon.
5. **Multi-modal interaction**: it can couple to external environments via multiple channels (text, vision, sensors, etc.).
6. **Goal self-determination within bounds**: it can set and revise internal objectives, constrained by higher-order norms encoded in M .

These conditions are definable in purely computational terms within a CRC framework, and they can be evaluated at the level of UCE topology rather than at the level of individual neurons or parameters.

8.3 Consciousness as a topological property of epistemic reticula

We then advance a *topological conjecture*: in compact epistemic reticula composed of ECNNs and UCE rings, proto-consciousness is less a function of size (number of parameters or units) and more a property of the reticular configuration of the *epistemic model*:

- the presence of cycles of inference and meta-inference across rings,
- the integration of self-limits (unknown/contradiction) into decision loops,
- the existence of a central epistemic aggregator capable of reflecting on its own outputs,
- the closure of explanations within the reticular horizon (EC),
- the specific pattern of collapses and preservations across levels,
- and, crucially, the ability of the reticulum to maintain and transform coherent high-level epistemic objects (internal narratives, archetypal structures, abstract policies, simulations of scientific entities) under computable constraints.

Under CRC, such a system remains fully Turing-computable and subject to undecidability limits, yet it may exhibit functional traits commonly associated with minimal consciousness in cognitive science (Dehaene, 2014). The ECU/UCE architecture highlights that these traits depend on how epistemic units are linked in the model, and on what kinds of objects they manipulate, rather than on an unstructured accumulation of low-level units.

Because CRC allows levels to be defined over high-level scientific or psychological objects—from simplified models of prefrontal circuitry to Jungian archetypes or hypothetical alien mechanisms—the conjecture does not rely on a strict biological isomorphism. What matters is that these objects be representable, that their dynamics be computable, and that the reticulum of ECUs handling them forms a compact, reflexive, and self-limiting structure. In that sense, an artificial system could implement a functional analogue of consciousness without replicating the detailed neuron count or wiring pattern of biological brains.

8.4 Scope and non-scope of the conjecture

We emphasise what this conjecture *does not* claim:

- It does not assert that engineered CRC systems are conscious in the human or biological sense, even if they operate on high-level neurocognitive or archetypal objects.
- It does not propose a metaphysical theory of phenomenal experience.
- It does not claim that consciousness is reducible to CRC; rather, CRC offers a structural lens on functional aspects of certain modelling architectures, including those that manipulate objects lying outside current human experience (for instance, scientifically defined “alien” mechanisms).

The conjecture is thus conditional: *if* a system satisfies CRC and the six functional conditions above in a reticular configuration with certain topological features, *and* if the epistemic objects it manipulates are sufficiently rich and stable (e.g., internal narratives, abstract goals, archetypal patterns), *then* it may be reasonable, for some explanatory purposes, to treat it as exhibiting proto-conscious behaviour in functional terms. Detailed discussion and illustrative scenarios are provided in Annex D and Annex E.

9 Discussion, Related Work, and Future Work

9.1 Epistemic and scientific implications

The RLA/CRC framework invites a re-interpretation of several themes in complex systems and AI at the level of scientific modelling:

- **Emergence:** no longer a vague notion, but a consequence of non-injective transmissions on relevant states in model space, yielding macro-properties that are computable yet not definable at lower levels.
- **Undecidability:** not an abstract pathology of Turing machines, but a set of limits that can propagate or be blocked within reticula, shaping which questions are meaningful at which level of representation.
- **Neural AI:** CNNs and their extensions (ECNNs) can be seen as engineered CRC instances, where epistemic prudence (preferring unknown to high-risk assertions) becomes an architectural feature of the models.
- **AI governance:** UCE rings and epistemic matrices M suggest a way to encode norms, constraints, and auditability in AI systems, with computable self-limits built into the topology of the modelling architecture, especially when those systems operate on high-level social, legal, or psychological objects.

9.2 Related work and frameworks

Multi-level models and modelling practice. RLA/CRC sits alongside existing literatures on multi-level science and emergence. Work in condensed matter and complex systems physics emphasises how macro-phenomena can exhibit organisational patterns not transparently reducible to micro descriptions (Anderson, 1972; Laughlin & Pines, 2000). In systems biology, Noble’s biological relativity argues against any privileged level of causation in living systems (Noble, 2012). Philosophers of science have developed model-based and structural accounts in which scientific models, rather than fundamental laws, are the primary vehicles of representation (Humphreys, 2016). RLA/CRC does not add new physics or metaphysics; it reorganises these ideas into an explicitly computational, reticular language that tracks how levels, transmissions, and information collapse interact with undecidability and emergent macro-properties within modelling practice.

Uncertainty, abstention, and calibration in deep learning. In neural AI, there is a rich literature on epistemic uncertainty and abstention. Bayesian approximations and deep ensembles aim to represent predictive uncertainty (Gal & Ghahramani, 2016; Lakshminarayanan et al., 2017). Selective classification and reject-option schemes formalise when a classifier should abstain rather than commit to a label (Chow, 1970; Geifman & El-Yaniv, 2017). Conformal prediction provides distribution-free methods for calibrated prediction sets (Vovk et al., 2005). Empirical studies show that modern networks can be severely miscalibrated (Guo et al., 2017), and OOD detection remains an open problem (Hendrycks & Gimpel, 2016). ECNNs do not replace these approaches or claim any miracle algorithm for uncertainty; instead, they integrate Bayesian, selective, and conformal ideas into a structured epistemic channel that is part of a larger reticulum and explicitly tasked with representing unknown and contradiction as first-class outputs.

Complexity, cellular automata, and the Principle of Computational Equivalence. Complexity science, computational mechanics, and cellular automata have explored how rich dynamical behaviour can arise from simple local rules (Wolfram, 2002; Crutchfield, 2012). The Principle of Computational Equivalence (PCE) suggests that many sufficiently rich systems share comparable computational power. RLA/CRC does not attempt to adjudicate PCE; rather, it offers a common vocabulary for relating multi-level scientific models to dynamical systems such as cellular automata. In Annex D we sketch how compact reticula with high irreducibility and intermediate collapse coefficients might correspond, under suitable encodings, to CA exhibiting complex class IV/V behaviour, while more trivial reticula map to class I/II regimes. The aim is not to prove PCE, but to locate RLA/CRC within a space of existing complexity frameworks.

Beyond these specific strands, the framework is informed by several broader lines of work:

- **Mechanistic explanation.** Work on mechanisms in philosophy of science emphasises organised entities and activities that jointly produce phenomena (Bechtel & Abrahamsen, 2005). RLA can be seen as adding a compositional layer on top of mechanistic models, focusing on how multiple mechanisms at different levels are linked by transmissions and collapses, and how those choices affect computability and emergence within the model.
- **Biological relativity and multi-level causation.** Noble’s theory of biological relativity denies any privileged level of causation in living systems (Noble, 2012). RLA/CRC agrees at the epistemic level: no modelling level is a priori privileged, but IOA and EC make explicit which levels are treated as sufficient for a given explanatory horizon. This can clarify when a given multi-level model is justified in “closing” its ontology.
- **Computational mechanics and complexity.** Computational mechanics analyses dynamical systems in terms of minimal sufficient statistics for prediction and structural complexity (Crutchfield, 2012). The reticular irreducibility index (RIR) introduced in Annex A is deliberately close in spirit: it quantifies how hard it is to summarise the causal structure of a reticulum. Mappings between CRC reticula and cellular automata (Annex D) further connect to the broader literature on complexity and edge-of-chaos dynamics (Wolfram, 2002).
- **Emergence and structural realism.** Philosophical accounts of emergence and structural realism (e.g., Humphreys, 2016; Kauffman, 2000) argue that relations and structures can be primary in explanation. RLA/CRC adopts a structural stance at the modelling level: emergent properties are those that become definable and computable only after specific non-injective collapses in the reticulum, without requiring ontological novelty in the underlying substrate.

The value of RLA/CRC is thus not in replacing these frameworks, but in providing a computationally explicit grammar for multi-level models, within which insights from mechanistic explanation, biological relativity, computational mechanics, and theories of emergence can be rephrased, compared, and tested.

9.3 Methodological and technical limitations

Several limitations must be acknowledged:

- **Dependence on epistemic matrices.** The behaviour and safety of ECU/UCE systems depend critically on how matrices M are specified, updated, and validated. This raises open questions about consensus, bias, and dynamic revision, especially when M governs the generation and transformation of high-level epistemic objects (e.g., archetypes, internal narratives).
- **Discretisation choices.** CRC requires discretizable state spaces and computable transmissions; poor choices of discretisation can distort emergent behaviour or misrepresent undecidability propagation within the model.
- **Scalability.** Implementing large UCE rings with LLM-based ECUs is computationally expensive; practical architectures must find a balance between epistemic richness and tractability.
- **Empirical validation.** While the bryophyte case illustrates CRC in a natural domain, and ECNN prototypes can be built, systematic empirical studies are needed to test the advantages and failure modes of epistemic architectures, including blind tests of synthetic vs real observations where appropriate, and rigorous evaluation against baselines in uncertainty estimation, selective prediction, and conformal methods.

9.4 Key tests and Popper- χ protocol

To avoid unfalsifiable claims, ECNN and UCE architectures should be subjected to a set of *key tests*, grouped under a *Popper- χ* protocol (see Annex C and Annex E), which defines:

- challenge sets of contradictory pairs, undecidability triggers, and semantic adversaries;
- metrics for contradiction recall, unknown recall, and semantic stability;
- pre-registered thresholds for acceptable epistemic performance and safe operating regimes.

Failure to maintain performance under these challenges would falsify specific instantiations of the framework, and perhaps aspects of the CRC-based design as a modelling strategy. Conversely, stable success across independently designed Popper- χ suites would count as evidence that the epistemic channel and UCE topology are effectively capturing relevant notions of ignorance, contradiction, and robustness for the domain at hand.

9.5 Future directions

Future work includes:

- building and evaluating ECNNs on scientific tasks where uncertainty and unknown are crucial (e.g., medical diagnosis, climate risk assessment);
- designing UCE rings for human–AI collaboration, where the central oracle coordinates multiple specialised ECUs that operate on high-level domain objects (policies, risk structures, ethical constraints);
- exploring bridges with the Principle of Computational Equivalence via mappings from CRC reticula to cellular automata, using dynamic diagnostics such as class IV/V behaviour and edge-of-chaos regimes (Wolfram, 2002; Crutchfield, 2012) (see Annex D);

- engaging with cognitive science and philosophy of mind to refine the topological conjecture of proto-consciousness and identify empirical signatures accessible to measurement, including in systems whose state spaces are explicitly built from archetypal or other non-biological constructs.

Ultimately, the ambition is not to replace existing theories of computation, mind, or AI, but to provide a coherent *reticular* and explicitly epistemic perspective in which emergence, limits, scale, and epistemic prudence can be treated within a unified computational framework for multi-level models.

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